UNCLASSIFIED — CONTROLLED DISTRIBUTION (Independent Product; see Handling)

WP-2026-INFRA-01

MAKRAN COAST INFRASTRUCTURE CLUSTER: OSINT-GATED FINANCIAL ENFORCEMENT FRAMEWORK

POLICY RESEARCH PAPER: WP-2026-INFRA-01 (Rev. 5.4 — Final)

Independent Policy Research Group | February 2026

DISSEMINATION VERSION — Coordinates excluded (see Handling)

Field	Value
Classification	UNCLASSIFIED — CONTROLLED DISTRIBUTION (Independent Product; see Handling)
Version	Rev. 5.4 (Final)
Date	February 2026
Prepared by	Independent Policy Research Group
Distribution	DISSEMINATION version — coordinates excluded. Full version maintained separately.
Intended use	Enforcement analysis and administrative record formation for non-kinetic financial enforcement options. Not determinative of end-use.
Rev. 5.4 adds	planning→analysis language · Level 4 descriptive caveat · explicit gated line in Exec Summary · Appendix E (What Would Change My Mind) · all prior hygiene fixes retained

SOURCES AND METHODS

Element	Detail
Primary imagery	Google Earth (historical imagery function, free). Sentinel-2 (ESA Copernicus Open Access Hub, free). Planet Labs / Maxar (commercial, optional upgrade).
AIS data	Commercial AIS via MarineTraffic or VesselFinder (~$500–2,000/yr). 24-month historical pull. No classified access required.
Timeline methodology	Node 3 estimate (12–18 months to operational) derived from construction stage, tempo comparison to other nodes, and standard industrial benchmarks. Uncertainty range applies.
Confidence framework	OBSERVED: directly visible, high confidence, independently verifiable. ASSESSED: inferred from profile + context, moderate confidence. INFERRED: analytical judgment, lower confidence, requires corroboration before Level 4 action.
Limitations	Commercial imagery cannot assess interior contents. All function assessments are analytical hypotheses. This reflects only sources reviewed for this product — does not imply exhaustive collection. This product does not employ classified collection, privileged law enforcement data, or non-public partner reporting. Absence of a signal in this product should not be interpreted as absence in the underlying environment.

ETHICS AND GOVERNANCE STATEMENT

This product is limited to non-kinetic financial enforcement and compliance policy analysis. It avoids operational targeting guidance, component enumerations, and stepwise interdiction procedures. The framework is designed to reduce misidentification risk, prevent over-claiming from OSINT, and support lawful administrative record formation — with explicit evidentiary gates and mandatory collateral-risk review prior to any external filing.

RESEARCH QUESTION AND CONTRIBUTION

Research Question

How can reproducible commercial OSINT signals be converted into an evidence-gated, legally defensible escalation framework for non-kinetic financial enforcement — one that is simultaneously transparent enough for independent publication and structured enough for administrative record formation?

Gap in Existing Literature

Existing scholarship on sanctions design addresses the strategic and legal dimensions of economic pressure campaigns. Drezner's coercive statecraft framework establishes when sanctions are likely to produce behavioral change. Nephew's practitioner account details how sanctions architecture is built for maximum leverage. Zarate's analysis of Treasury-centered financial warfare describes the institutional evolution of financial enforcement as a strategic instrument.

None of these frameworks address the evidentiary translation problem at the operational level: how does an analyst move from a commercial imagery observation to an administratively staffable enforcement record without classified collection, without overclaiming from OSINT, and without producing a product that counsel will reject for predicate deficiency? This paper addresses that gap.

Contribution

This paper contributes a replicable OSINT-to-enforcement translation methodology to the literature on financial enforcement design. Specifically, it demonstrates that commercial geospatial intelligence, structured through explicit evidentiary gates and identity resolution standards, can produce an administratively defensible enforcement scaffold without classified sourcing. The framework is falsifiable — each assessment carries explicit disconfirming indicators — and reproducible — all imagery and AIS data sources are identified with query parameters sufficient for independent replication.

Scope and Limitations of Claim

This paper does not claim to establish end-use of any identified facility, to provide actionable intelligence, or to substitute for classified collection. The contribution is methodological: a framework for converting OSINT observations into evidence-gated enforcement scaffolding. The Makran coast case study is an application of the methodology, not the methodology's primary claim.

Key References

Reference	Relevance to This Framework
Drezner, D.W. (2011). "Sanctions Sometimes Smart: Targeted Sanctions in Theory and Practice." International Studies Review 13(1).	Establishes conditions under which targeted economic pressure produces behavioral change. Informs escalation gate logic.
Nephew, R. (2017). The Art of Sanctions: A View from the Field. Columbia University Press.	Practitioner framework for sanctions architecture and leverage design. Informs chokepoint mapping and SPAD methodology.
Zarate, J.C. (2013). Treasury's War: The Unleashing of a New Era of Financial Warfare. PublicAffairs.	Documents institutional evolution of financial enforcement. Provides practitioner context for Section 311 and E.O. authority application.
Albright, D. & Stricker, A. (2014). "Revisiting South Africa's Nuclear Weapons Program." Institute for Science and International Security.	Methodological precedent for OSINT-based infrastructure assessment with explicit confidence rubrics.
Financial Action Task Force (2022). Guidance on Proliferation Financing Risk Assessment and Mitigation.	Establishes international typology standards for proliferation financing administrative records. Informs custody hook framework.

0. CASE OPENING SUMMARY

0.1 Purpose

This paper examines how reproducible OSINT signals can be converted into an evidence-gated, legally defensible escalation framework for non-kinetic financial enforcement. It presents a case-opening and evidence-building referral for enforcement options against an integrated infrastructure cluster on Iran's Makran coast. Designed to (i) establish a reproducible OSINT baseline and (ii) initiate Level 1–3 enforcement analysis under existing authorities, while explicitly gating Level 4 actions pending corroboration beyond commercial imagery. This product supports prioritization, tracing initiation, and administrative record formation; it is not determinative of end-use and not sufficient for high-impact actions absent corroboration.

0.2 Establishes / Does Not Establish

ESTABLISHES: Integrated industrial cluster under active construction; node-level mapping to statutory tools; finite pre-operational window for certain nodes.

DOES NOT ESTABLISH: End-use of Node 3; contents of Node 4; confirmed dual-use or military linkage. Level 4 designations require corroboration beyond imagery.

0.3 Evidence Level at Referral

Level	Status	Actions Available	Timeline
Level 1	CURRENT — commercial imagery baseline	Commercial AIS pull for Node 7/9	Now — 30 days
Level 2	TARGET — imagery + AIS vessel ID	§ 314(a) requests; shipping record analysis	30–60 days
Level 3	TARGET — imagery + AIS + financial network ID	Section 311 admin record; Entity List; CISADA	90–180 days
Level 4	GATED — function confirmation required	Additional collection beyond imagery. Financial enforcement proceeds in parallel.	Collection-dependent

0.4 Priority Nodes

Node 3 (large unfinished industrial complex) — CRITICAL PRIORITY. Estimated 12–18 months from operational. Node 7 (deep-water port) — CRITICAL PRIORITY. Node 9 (fuel refinery) — HIGH PRIORITY.

0.5 Requested Actions — Structured for Staffing Speed

Request 1 — Level 1 (30 days):

Action	Lead	Support	Output
Commercial AIS pull (24-month) Node 7/9 — vessels, flags, owners, P&I	Interagency analytic cell (Treasury lead)	FinCEN, OFAC, BIS	AIS vessel list + ownership chains
Sentinel-2 imagery baseline all nodes — archive + current	Interagency analytic cell	All	Imagery baseline per node
Internal BIS/OFAC preliminary review	BIS; OFAC (parallel)	IC coordination	Entity flags; preliminary eligibility assessment

Request 2 — Level 2 (30–60 days):

Action	Lead	Support	Output
§ 314(a) requests keyed to AIS vessel ownership and shipping record analysis	FinCEN	Treasury TFFC	Financial institution responses; entity linkage
Supplier identity resolution per Annex C standards	BIS	OFAC; DOJ	Identity packets ready for Notice Letters
Registry de-flagging notifications (non-binding) for AIS-confirmed vessels	State (maritime liaison)	Treasury OFAC	Flag state engagement record

0.6 Handling

Coordinates excluded from this version. Controlled version (with Annex A) maintained separately under restricted distribution.

0.7 Limitations and Confidence Summary

All node function assessments are analytical hypotheses based on physical profile, construction pattern, and strategic context. OSINT cannot confirm end-use. This statement reflects only sources reviewed and does not imply exhaustive collection. Absence of evidence in this product does not imply absence in the environment.

0.8 Intended Use and Non-Use

INTENDED: Supports prioritization, tracing initiation, and administrative record formation for non-kinetic financial enforcement analysis options. NOT INTENDED: Determinative of end-use. Not sufficient for high-impact actions absent corroboration. Not a substitute for classified collection or law enforcement data. Not for kinetic or operational targeting.

1. EXECUTIVE SUMMARY

Key Judgments

#	Judgment	Confidence	Basis
KJ-1	An integrated infrastructure cluster comprising ten discrete facilities is under active construction on Iran's Makran coast.	OBSERVED — high confidence	Commercial satellite imagery (Google Earth, Sentinel-2) through Oct–Nov 2025. Independently verifiable.
KJ-2	The cluster's engineering design — self-sufficient power, water, fuel, port, and communications — is consistent with a resilience-oriented logistics enablement posture. End-use is unestablished.	ASSESSED — moderate confidence	Based on observed self-sufficiency features and coordinated construction sequencing. Does not rest on direct observation of facility contents.
KJ-3	Node 3 (large unfinished industrial complex) is estimated at 12–18 months from operational status. Supply chain constraint options are substantially more available before operational status than after.	ASSESSED — moderate confidence, uncertainty range applies	Construction stage visible in imagery; tempo comparison to other nodes; standard industrial benchmarks.
KJ-4	Level 1–3 financial enforcement options are eligible for consideration under existing statutory authority without additional collection beyond commercial OSINT.	OBSERVED (legal authorities) / ASSESSED (predicate adequacy)	Statutory authorities publicly codified. Predicate adequacy assessed based on current evidence level.
KJ-5	Level 4 actions (E.O. 13382 facility designation, § 2339D referral) are explicitly gated. The escalation gate for Level 4 is not satisfied by commercial imagery alone.	GATED — additional collection required	Physical profile and construction pattern do not constitute function confirmation.

Commercial satellite imagery has identified a self-sufficient industrial complex under active construction on Iran's Makran coast, Sistan-Baluchestan Province. Ten discrete facilities within approximately 100 square miles provide independent power, desalinated water, fuel refining, hardened storage, deep-water port access, and protected communications. The complex faces the open Indian Ocean outside Strait of Hormuz transit, approximately 70–80 kilometers from CPEC terminus at Gwadar, Pakistan.

The construction pattern and integrated engineering design are ASSESSED (moderate confidence) as consistent with a resilience-oriented logistics and industrial enablement posture, based on observed self-sufficiency features — dedicated power, desalination, fuel capability, port access, and protected communications — and their coordinated construction sequencing. End-use remains unestablished, and this assessment does not rest on direct observation of facility contents. Escalation to Level 4 actions is explicitly gated on corroboration beyond commercial imagery. Each node maps to specific enforcement options under existing statutory authority, with explicit escalation gates and decision authority. Section IX provides an illustrative implementation checklist. End-use is unestablished; escalation remains gated.

1.1 Competing Hypotheses and Disconfirming Indicators

This product evaluates multiple plausible explanations for the observed infrastructure pattern. The current assessment reflects the best-supported hypothesis given reviewed OSINT; it remains revisable upon corroboration of disconfirming indicators.

Hypothesis	Explanation	Would Be Strengthened By	Would Be Weakened By	Current Weight
H1: Civilian economic development (baseline)	Industrial and port expansion consistent with regional development initiatives for an underdeveloped province.	Verified civilian permitting; publicly documented commercial operators; routine civilian logistics profile.	Consistent association with designated entities; repeated covert logistics patterns; verified controlled-goods supply chains.	POSSIBLE — not excluded; insufficient evidence to confirm or exclude
H2: Resilience-oriented logistics enablement (current assessment; moderate confidence)	Engineered self-sufficiency consistent with continuity-of-operations logistics posture.	Documentary evidence of state or security-linked operators; sanctioned maritime networks; financial intermediaries consistent with evasion typologies.	Transparent commercial counterparties; conventional market-facing operations; absence of designated-entity nexus after identity resolution.	ASSESSED — moderate confidence; best-supported by observed construction pattern and sequencing
H3: Dual-use or military-relevant enablement (explicitly gated; not established)	Potential dual-use relevance cannot be excluded from physical profile alone.	Corroborated end-use indicators beyond imagery; controlled procurement records; admissible financial or documentary nexus meeting Level 4 gate.	Affirmative civilian end-use confirmation; neutral counterparties satisfying identity standards.	INFERRED — lower confidence; Level 4 escalation gate not yet satisfied

2. STRATEGIC CONTEXT

2.1 Geographic Logic

Iran's Persian Gulf infrastructure is constrained by Strait of Hormuz transit. Makran coast provides direct Indian Ocean access. Vessels operating here reach the Arabian Sea, Gulf of Oman, East Africa, and South Asia without Strait passage — consistent with infrastructure designed for Strait disruption continuity.

2.2 CPEC Adjacency (INFERRED — requires financial intelligence corroboration)

CPEC terminus at Gwadar is 70–80 kilometers east. Transactions between this complex and CPEC-linked entities may involve Chinese state-affiliated intermediaries subject to CAATSA (22 U.S.C. § 9241) and CISADA (22 U.S.C. § 8513). INFERRED — do not apply secondary sanctions on this basis alone.

2.3 Construction Sequence (ASSESSED — moderate confidence)

Sentinel-2 2022–2025 archive shows: utilities first, hardened storage second, port and communications third, large industrial capacity last. Consistent with planned integrated development. Confidence moderate — based on imagery change detection, not confirmed construction records.

3. ANALYTIC CONFIDENCE FRAMEWORK

Level	Definition	Basis	Enforcement Implication
OBSERVED	Directly visible in commercial imagery. Physical structure, dimensions, construction activity.	Google Earth, Sentinel-2	Supports Level 1–2 actions. Independently verifiable.
ASSESSED	Inferred from physical profile, construction pattern, strategic context.	Physical profile + strategic context	Supports Level 2–3 with corroboration. Insufficient alone for E.O. 13382 designation.
INFERRED	Analytical judgment from pattern and context. Not supported by imagery alone.	Strategic analysis	Level 4 escalation gate requires independent corroboration. Do not assert military/dual-use on INFERRED alone.

3.1 Defined Terms

Term	Definition as Used in This Framework
SLRI	Satellite Logistics Readiness Indicators — observable OSINT metrics used to assess cluster operational status and enforcement effectiveness over time (Section XI).
SPAD	Spare Parts and Dependence — supply chain reliability de-risking doctrine targeting high-consumable or high-maintenance items that cannot be easily domesticated (CONTAIN-01).
Custody hook	A document category that connects an identified entity to the evidentiary record. Presence of a custody hook supports nexus analysis but does not independently establish illicit activity.
Nexus	A documented relationship between an identified entity and the enforcement subject, established by one of: control (decision-making authority), benefit (economic benefit from

	operations), ownership (equity or equivalent interest), or agency (acting on behalf of subject in transactions). Documentary basis required — see Annex C. Geographic proximity alone does not constitute nexus.
Escalation gate / Gate	The minimum evidentiary predicate that must be satisfied before a specific enforcement option is eligible for consideration. Satisfaction of the gate does not automatically trigger the action.
Identity packet	The completed entity identification dossier satisfying Pull-Out 3 and Annex C standards, required before any Notice Letter, Entity List addition, or designation package is submitted.

4. FACILITY NODE INVENTORY

Generalized location references only. Precise coordinates in Controlled Annex A only. Each entry: Observed → Assessed → Enforcement mapping with escalation gate standards. Do not assert military/dual-use linkage based solely on physical profile (requires Level 4 corroboration).

NODE 1 — Power Generation Facility

General location: Northern sector, inland

OBSERVED

Observed: Large power generation facility: turbine hall, fuel storage, switchyard, transmission infrastructure. Capacity appears to exceed local residential need.

ASSESSED

Assessed function: Primary power source for cluster. Designed for independent operation disconnected from Iranian national grid. Redundant fuel storage consistent with sustained isolated operations.

Primary SPAD reliability target — turbine maintenance components, control systems represent supply chains that cannot be easily domesticated.

Action	Authority	Minimum Predicate	Framework Ref
SPAD reliability targeting: turbine maintenance, control systems, fuel supply chain	E.O. 13382; 15 C.F.R. § 744.21	Operator designated under E.O. 13382 AND specific vendors identified via AIS/shipping. Both predicates required.	CONTAIN-01 SPAD; SIEGE-01 CP4
SLRI monitoring: power output as operational readiness indicator	CONTAIN-01 SLRI	Monitoring only. No escalation gate.	CONTAIN-01

NODE 2 — Water Treatment / Desalination Plant

General location: Northern sector, adjacent to Node 1

OBSERVED

Observed: Desalination plant with intake pipelines from coastline, storage tanks, distribution infrastructure. Scale consistent with industrial process water supply.

ASSESSED

Assessed function: Primary potable and process water source for cluster. Desalination indicates design for independence from regional water infrastructure.

SPAD reliability target — membrane replacement, chemical treatment supplies represent consumable supply chains.

Action	Authority	Minimum Predicate	Framework Ref
SPAD reliability targeting: desalination membranes, chemical treatment supplies	E.O. 13382; 15 C.F.R. § 744.21	Operator designated AND specific vendors identified. Both required.	CONTAIN-01 SPAD

NODE 3 — Large Unfinished Industrial Complex ⚠️ PRIORITY / PRE-OPERATIONAL WINDOW

General location: Central sector, north of hardened perimeter

OBSERVED

Observed: Large-footprint construction: foundation complete, structural steel framing in progress, early enclosure. Active construction Oct–Nov 2025 Sentinel-2 imagery. Footprint scale and siting are atypical for standard civilian industrial norms in this province (ASSESSED; moderate confidence).

ASSESSED

Assessed function: Industrial production or large-system storage. Scale, isolation, and proximity to hardened storage and deep-water port are consistent with heavy manufacturing, large-scale assembly, or storage of large systems. Specific end-use NOT established by commercial imagery. Assessed moderate confidence. Disconfirming indicator: documented civilian permitting and routine commercial counterparties consistent with benign industrial development would revise this assessment downward.

INFERRED

Inferred: Cluster logic — adjacency to Node 4, Node 7, Node 9 — consistent with a production-and-logistics design. Requires corroboration. Do not assert military or dual-use function on this inference alone.

PRIORITY: Estimated 12–18 months from operational (uncertainty range applies). Supply chain constraint before operational status is substantially more achievable than capability de-risking after. Window currently open.

Action	Authority	Minimum Predicate	Framework Ref
BIS Entity List review for confirmed contractors/suppliers	15 C.F.R. Part 744; 50 U.S.C. § 4819	Contractor identity confirmed via corporate registry, shipping manifests, or export license records. Not physical profile alone.	SIEGE-01 CP4; CPF-02
Notice Letters to active construction suppliers	50 U.S.C. § 4819(c); 18 U.S.C. § 2339D	Supplier identity confirmed per Annex C AND end-user designation confirmed. Both predicates required.	UNIFIED-01 §4.1
Section 311 finding / special measures for construction financing institutions	31 U.S.C. § 5318A	Institution identified via SAR or § 314(a) AND § 5318A(c) threshold met AND admin record complete AND Federal Register notice prepared.	SIEGE-01 CP1; CPF-02
E.O. 13382 designation (blocking upon designation) — GATED at Level 4	E.O. 13382; 50 U.S.C. § 1702	Function corroboration required beyond physical profile. Do not designate on imagery alone. Level 4 escalation gate not yet satisfied.	CPF-02 — Level 4 gate

NODE 4 — Hardened Secure Industrial Perimeter

General location: Central sector, east of Node 3

OBSERVED

Observed: Enclosed compound with perimeter wall, two large primary structures, controlled access. Standoff distances and perimeter construction profile are atypical for standard civilian warehousing norms in this province (ASSESSED; moderate confidence).

ASSESSED

Assessed function: Secure storage of high-value or sensitive equipment. Hardened perimeter consistent with materials requiring physical access control and environmental protection. Contents NOT observable and NOT established by commercial imagery. Disconfirming indicator: affirmative civilian permitting records or transparent commercial operator documentation would revise this assessment downward.

Do not assert military or WMD storage based on hardening profile alone. Level 4 corroboration required before E.O. 13382 designation.

Action	Authority	Minimum Predicate	Framework Ref

E.O. 13382 designation (blocking upon designation) — GATED at Level 4	E.O. 13382; 50 U.S.C. § 1702	Contents or function corroborated by additional collection beyond imagery. Physical hardening alone does not satisfy escalation gate.	CPF-02 — Level 4 gate
Maritime enforcement coordination for vessels delivering to adjacent port	E.O. 13382; 46 U.S.C. § 70201	Cargo manifest indicates controlled or dual-use goods AND vessel associated with designated entity. Jurisdiction, flag-state cooperation, and applicable legal process required.	SIEGE-01 CP7; CPF-02 §5
SLRI tracking: facility activation as cluster readiness milestone	CONTAIN-01 SLRI	Monitoring only. No escalation gate.	CONTAIN-01

NODES 5 & 6 — Ocean Water Intake Pump Houses (Redundant Pair)

General location: Coastline, southern sector

OBSERVED

Observed: Two pump house structures at coastline with intake infrastructure and pipeline routing northward. Dual installation indicates redundant intake capacity.

ASSESSED

Assessed function: Raw seawater intake for desalination (Node 2) and cooling water for Nodes 1 and 3. Redundant design confirms engineering for operational resilience.

Primary SPAD reliability targets — pump maintenance components, anti-fouling treatments represent consumable supply chains.

Action	Authority	Minimum Predicate	Framework Ref
SPAD reliability targeting: pump maintenance, marine engineering services	E.O. 13382; 15 C.F.R. § 744.21	Operator designated AND specific vendors identified. Both predicates required.	CONTAIN-01 SPAD; SIEGE-01 CP4

NODE 7 — New Deep-Water Port Construction ⚠️ PRIORITY NODE

General location: Eastern sector, coastline

OBSERVED

Observed: Active port construction: pier/quay development, breakwater construction, dredging operations, landside logistics infrastructure. Pier scale indicates design for large vessels.

ASSESSED

Assessed function: Primary maritime logistics interface. Enables inbound heavy equipment delivery, outbound product movement. Physical terminus of shadow logistics chain.

AIS monitoring priority — vessel calls establish the evidentiary foundation for financial enforcement actions.

Action	Authority	Minimum Predicate	Framework Ref
Commercial AIS 24-month pull — vessel ID, flag registry, P&I club, ownership chains	CONTAIN-01 SLRI	Commercial AIS subscription active. Generalized bounding boxes (see Handling).	SIEGE-01 CP7
Registry de-flagging notifications (non-binding)	E.O. 13382	AIS confirms regular vessel calls AND vessel/operator linked to designated entity or documented illicit cargo. AIS evidence required.	SIEGE-01 CP7; CPF-02 §4.3
§ 981(k) asset substitution referral	18 U.S.C. § 981(k)	Foreign bank + U.S. correspondent account confirmed + probable cause established + court order required. All four required.	CPF-02 §5.2

NODE 8 — Protected Communications Facility

General location: Eastern sector, inland from Node 7

OBSERVED

Observed: Communications facility with hardened structures, antenna arrays, protected cable routing. Physical profile consistent with encrypted or redundant communications infrastructure.

ASSESSED

Assessed function: Protected communications node for cluster. Consistent with infrastructure designed to maintain command/coordination during external pressure.

SPAD target for communications systems. Secondary enforcement priority behind Nodes 3, 7, 9.

Action	Authority	Minimum Predicate	Framework Ref
SPAD targeting: communications equipment supply chain	E.O. 13382; 15 C.F.R. § 744.21	Operator designated AND specific vendors identified.	CONTAIN-01 SPAD
SLRI secondary monitoring node	CONTAIN-01 SLRI	Monitoring only. No escalation gate.	CONTAIN-01

NODE 9 — Fuel Refinery and Secondary Port ⚠️ PRIORITY NODE

General location: Western sector, coastline

OBSERVED

Observed: Refinery infrastructure with storage tanks, process equipment, flaring infrastructure, and secondary port facility enabling direct fuel loading to vessels.

ASSESSED

Assessed function: Fuel refining and distribution for cluster and regional maritime operations. Secondary port enables direct vessel bunkering. Consistent with petroleum products storage and distribution.

INFERRED

Inferred: Potential CISADA petroleum transaction nexus — transactions involving petroleum products from this facility may involve entities subject to CISADA (22 U.S.C. § 8513). INFERRED — requires transaction-level corroboration.

CISADA petroleum transaction monitoring is the primary financial intelligence hook for this node.

Action	Authority	Minimum Predicate	Framework Ref
CISADA 180-day warning letters for petroleum transactions	22 U.S.C. § 8513	Transaction records confirming petroleum product purchases from or through Node 9 facility operators. Entity identity per Annex C.	SIEGE-01 CP9; CPF-02 §6
SLRI primary monitoring: port call frequency, vessel size, cargo patterns	CONTAIN-01 SLRI	Monitoring only. AIS commercially available.	CONTAIN-01

NODE 10 — Inland Oil / Power Generating Site

General location: Northern sector, inland

OBSERVED

Observed: Inland facility with storage tanks, power generation infrastructure, access road.

ASSESSED

Assessed function: Secondary power generation and fuel storage. Consistent with backup power and fuel reserve for cluster independence.

Lower enforcement priority. Monitor for supply chain connections to Nodes 1 and 9.

Action	Authority	Minimum Predicate	Framework Ref
SPAD monitoring: fuel supply chain	CONTAIN-01 SLRI	Monitoring only. No escalation gate.	CONTAIN-01

4.5 ENFORCEMENT DECISION TREE

Evidence Level	Unlocked Options	Escalation Gate (Minimum Predicate)	Decision Owner	Timebox
Level 1 Current status OSINT patterns only	Sentinel-2 baseline AIS pull Internal BIS/OFAC review SLRI monitoring	None — conceptually feasible within standard review processes	Interagency analytic cell (Treasury lead)	Now–30 days
Level 2 Imagery + AIS vessel ID (30–60 days)	Registry de-flagging notifications Notice Letters § 314(a) requests FATF coordination	Notice Letters: supplier identity per Annex C AND end-user designation confirmed (both required). Registry: AIS confirms regular calls AND vessel linked to designated entity or illicit cargo.	BIS (Notice Letters) State/Treasury (registry) FinCEN (§314(a))	30–60 days
Level 3 Imagery + AIS + financial ID (90–180 days)	Section 311 finding / special measures § 981(k) asset substitution referral Entity List additions CISADA warnings	311: institution ID via SAR/§ 314(a); § 5318A(c) threshold met; admin record complete. § 981(k): foreign bank + U.S. correspondent confirmed; probable cause established. Entity List: credible evidence; identity per Annex C.	FinCEN (311) DOJ MLARS (§ 981(k)) BIS/OFAC (Entity List)	90–180 days
Level 4 Imagery + AIS + financial + function corroboration (GATED)	E.O. 13382 facility designation § 2339D material support prosecution Maritime enforcement coordination Escalated cross-framework coordination	Level 4 entries describe policy lanes at a high level; this product does not provide operational procedures. Function corroborated beyond physical profile. End-use and contents confirmed. Operator identity established. Do not assert military/dual-use on profile alone. Note: Level 3 financial enforcement proceeds in parallel — does not require Level 4.	Relevant financial enforcement authority Relevant legal authority Relevant maritime enforcement authorities	Collection-dependent
Escalation SLRI shows continued activity despite Levels 1–3	CISADA layering for Node 9 petroleum transactions Quarterly SLRI reassessment	Corroborated transaction patterns showing continued petroleum activity at Node 9 despite Level 3 pressure.	Financial enforcement function (OFAC)	Quarterly reassessment

5. CLUSTER ANALYSIS AND CHOKEPOINT MAPPING

5.1 Systems Interdependence

The ten nodes form an integrated system. Node 1 (power) feeds Nodes 2, 3, 4, 8. Node 2 (water) feeds Nodes 1, 3. Node 9 (fuel) feeds Nodes 1, 7 (bunkering). Node 7 (port) serves as the primary logistics interface for all inbound supply. Disruption of any single node degrades cluster-wide capability — supply chain constraint on Node 3 construction materials or Node 7 vessel calls has compounding effects.

5.2 SIEGE-01 Chokepoint Mapping

Chokepoint	Node(s)	Enforcement Lane	SPAD Focus
CP1 — Construction financing	Node 3	Section 311 / § 5318A	Construction material supply chains
CP4 — Equipment supply chains	Nodes 1, 2, 3, 4, 5/6	BIS Entity List; Notice Letters	Turbines, membranes, control systems
CP7 — Maritime logistics	Node 7	AIS monitoring; registry; § 981(k)	Vessel bunkering, cargo manifests
CP9 — Petroleum transactions	Node 9	CISADA 180-day warnings	Petroleum product transactions

5.3 SPAD Reliability Degradation Targets

Spare Parts and Dependence (SPAD) targeting focuses on high-consumable or high-maintenance items that cannot be easily domesticated:

Node / System	SPAD Target Items
Power generation (Node 1)	Gas turbine components, generator parts, control system firmware, cooling system components
Desalination (Node 2)	Reverse osmosis membranes (18–24 month cycles), anti-scalant chemicals, high-pressure pump components
Industrial complex (Node 3)	Specialized manufacturing equipment, precision components, industrial control systems
Refinery (Node 9)	Catalyst replacement (18–24 month cycles), inspection services, process control systems
Port (Node 7)	Crane components, dredging equipment, navigation systems, mooring hardware

6. SLRI OBSERVABILITY PROTOCOL

Satellite Logistics Readiness Indicators (SLRI) — four observable metrics for tracking cluster status and enforcement effectiveness:

Indicator	Source	Cost	What It Tracks	Cadence
Port call frequency + vessel size at Node 7	AIS (MarineTraffic, VesselFinder)	~$500–2,000/yr	Supply throughput; shadow fleet vessel ID	Weekly
Construction activity at Node 3	Sentinel-2 (free) or Planet/Maxar	Free–$1,500/mo	Timeline to operational capability	Monthly
Refinery flaring signature at Node 9	Sentinel-2 infrared (free)	Free	Operational tempo; fuel production activity	Weekly
Facility activation indicators (all nodes)	Sentinel-2 change detection	Free	Workforce presence; vehicle activity; power	Monthly

7. CONSOLIDATED LEGAL AUTHORITY REFERENCE

Authority	Citation	Application to This Framework	Applicable Nodes
E.O. 13382 Designation	E.O. 13382; 50 U.S.C. § 1702	Blocking designation for WMD proliferators and supporters. Level 4 gate — requires function corroboration beyond imagery.	Nodes 3, 4 (Level 4 only)
BIS Entity List	15 C.F.R. Part 744; 50 U.S.C. § 4819	Export restriction for entities posing national security concern. Requires credible evidence — not physical profile alone.	Node 3 (suppliers)
Notice Letters	50 U.S.C. § 4819(c); 18 U.S.C. § 2339D	Formal notice establishing knowledge element. Requires supplier identity per Annex C AND end-user designation confirmed.	Node 3 (suppliers)
Section 311 Finding / Special Measures	31 U.S.C. § 5318A	Special measures against primary money laundering concern. Requires institution identification via SAR/§ 314(a), § 5318A(c) factors, complete admin record, Federal Register notice.	Node 3 (financing institutions)
§ 981(k) Asset Substitution	18 U.S.C. § 981(k)	Civil forfeiture via U.S. correspondent accounts. Requires foreign bank + U.S. correspondent confirmed + probable cause + court order. All four required.	Node 7 (maritime finance)
CISADA	22 U.S.C. § 8513	Secondary sanctions for petroleum transactions. Requires corroborated transaction records — not inference from proximity.	Node 9
CAATSA	22 U.S.C. § 9241	Secondary sanctions for significant transactions with Iranian defense sector. CPEC-adjacent — INFERRED; requires financial intelligence corroboration.	CPEC adjacency (INFERRED)
FDPR	15 C.F.R. § 734.9	Foreign Direct Product Rule — reaches foreign-made items using U.S. technology. Requires Entity List addition in place first.	Node 3 (foreign suppliers)
Material Support (WMD)	18 U.S.C. § 2339D	Expert advice, financing, or material support to WMD program. WMD function must be confirmed (Level 4). Notice Letter establishes knowledge element.	Suppliers (Level 4 confirmation)

8. CONCLUSION

8.1 Establishes

From commercially available OSINT: (1) a self-sufficient infrastructure cluster is under active construction on Iran's Makran coast; (2) construction pattern and integrated engineering are ASSESSED (moderate confidence) as consistent with a resilience-oriented logistics and industrial enablement posture — end-use remains unestablished; (3) each node maps to specific enforcement options under existing statutory authority with explicit escalation gates and decision authority; (4) the pre-operational window for Node 3 is currently open.

8.2 Does Not Establish

Specific end-use of Node 3 or contents of Node 4. All function assessments are analytical hypotheses. E.O. 13382 designations require Level 4 corroboration. Financial enforcement (Section 311 finding / special measures, § 981(k)) can proceed at Level 3 without Level 4 function confirmation.

8.3 Recommended Next Step (Independent Analytic Judgment)

Level 1 actions are low-cost, reversible, and primarily evidentiary: they build a reproducible baseline and administrative record foundation for any later consideration of higher-impact tools. The Sentinel-2 baseline and commercial AIS pull are therefore the recommended first moves for any stakeholder evaluating this cluster within the framework's evidentiary gates.

Node 3 is estimated at 12–18 months from operational status (uncertainty range applies). Supply chain constraint before operational status is substantially more achievable than capability de-risking after. The pre-operational window is currently open.

8.4 FAILURE MODES AND THREATS TO VALIDITY

This section stress-tests the framework itself. A senior practitioner or academic reviewer should ask not only whether the analysis is correct, but whether the methodology could fail in ways that produce confident-looking but incorrect outputs.

Failure Mode	Mechanism	Why It Matters	Mitigation in This Framework	Residual Risk
OSINT-to-intent leap	Analyst infers intent or end-use from physical profile alone.	Produces high-confidence-looking assessments that cannot survive legal challenge or peer review.	Explicit three-tier confidence rubric. Level 4 gate requires function corroboration beyond imagery. Competing hypotheses with H1 civilian baseline.	Moderate. Physical profile assessments remain hypothesis-level.
False entity resolution	Two distinct entities assigned the same identity due to name variants, shell company structures, or address reuse.	Produces incorrect enforcement targets. Top failure mode in AML/sanctions identity work.	Annex C identity standards require legal entity name, jurisdiction, physical address, beneficial ownership, and independent transaction record.	Moderate-high. Framework defines the standard but cannot perform the resolution.
Construction timeline overconfidence	Node 3 operational timeline estimate (12–18 months) presented as more precise than methodology supports.	Creates false urgency, drives premature escalation, damages credibility when inaccurate.	Explicit "uncertainty range applies" at every timeline mention. Methodology documented. Quarterly SLRI review updates	Low-moderate. Readers may anchor on point estimate regardless.

			estimate.	
Evidentiary gate drift	Analysts treat the gating framework as a formality and advance actions before predicates are genuinely satisfied.	Undermines the framework's primary purpose — preventing premature escalation.	Quarterly escalation review checklist. Downshift standards provide off-ramp.	Moderate. Institutional pressure can override process discipline.
OSINT availability bias	Facilities visible in commercial imagery are assessed; those not visible are not. May produce systematically incomplete picture.	Analysts may draw cluster-level conclusions from a node inventory selected by visibility, not strategic significance.	Methods section explicitly states: "does not imply exhaustive collection." Absence of signal ≠ absence in environment.	Moderate-high. Cannot be eliminated without additional collection.
Legal authority misapplication	Authorities cited without satisfying the statutory predicate.	Renders enforcement action legally vulnerable.	Every authority includes explicit escalation gate stating minimum predicate. Legal hygiene pass at Rev. 5.2–5.3.	Low. Framework is predicate-explicit. Risk is in execution, not design.

The most important residual risk is the one this framework cannot fully mitigate: the gap between OSINT-based assessment and confirmed fact. This framework defines what evidence is needed to close that gap, structures how to pursue it, and prevents premature escalation across it. It does not close the gap itself.

IX. IMPLEMENTATION CHECKLIST (ILLUSTRATIVE)

Ten illustrative elements, each specifying what it enables, the minimum predicate, and the expected output artifact. Elements are sequenced to reflect evidentiary dependency — each builds the documentary foundation for the next. Elements 1–4 are conceptually feasible within standard agency review processes.

#	Element	What It Enables	Minimum Predicate	Output Artifact
1	Sentinel-2 imagery baseline pull (Copernicus Open Access Hub — free)	SLRI change detection baseline; Node 3 construction timeline monitoring; reproducible imagery record for administrative record formation	None. Conceptually feasible within standard review processes.	Dated imagery archive per node, stored with access log.
2	Commercial AIS 24-month data pull (Node 7 + Node 9 generalized bounding boxes; see Handling)	Vessel ID; flag registry; beneficial ownership chains; P&I club affiliations; behavioral pattern	Commercial subscription (~$500–2,000). Generalized bounding boxes (coordinates in Controlled version only).	AIS vessel registry per node. Flag/ownership/P&I chains. Behavioral baseline.
3	Internal BIS/OFAC preliminary review (no external action)	Preliminary entity flags; legal authority mapping; enforcement analysis readiness	None. Internal review only.	Internal flags memo. Entity list cross-reference.
4	FinCEN § 314(a) information sharing requests (keyed to AIS/shipping findings)	Financial institution identification; SAR pattern analysis; Section 311 predicate building	Entity identifiers from AIS or shipping records. § 314(a) standard criteria.	§ 314(a) responses. SAR patterns. Institution identification.
5	Supplier identity packet per Annex C standards	Notice Letter predicate; Entity List submission predicate; admin record foundation	Entity name confirmed via AIS, shipping manifest, or export license records.	Identity packet per supplier. Filed against Annex C checklist.
6	Annex B documentary hooks tracing plan	Admin record structure; custody chain for all evidentiary claims	Identity packet complete (Element 5).	Documentary hooks map. Custody log initiated.
7	Notice Letter package (BIS, per Annex C)	Knowledge element for § 2339D; supply chain constraint; admin record	Supplier identity confirmed per Annex C AND end-user designation confirmed. Both required.	Delivered Notice Letters with delivery confirmation. Admin record entry.
8	Section 311 admin record skeleton (Federal Register posture)	Section 311 finding / special measures predicate	Institution identified via § 314(a)/SAR; § 5318A(c) factors documented; admin record complete; Federal Register notice prepared.	Complete § 5318A admin record. Federal Register notice draft.
9	Parallel BIS + OFAC sequencing (avoid serial enforcement latency)	Simultaneous Entity List + designation package development	Elements 5–8 complete. Interagency coordination initiated.	Coordinated enforcement package. Sequenced action timeline.
10	Downshift guardrail — collateral risk	Prevents over-escalation; maintains	Required before any external filing.	Collateral risk assessment. GL

	review before any external filing	framework integrity; protects admin record	Quarterly escalation gate review.	scope confirmation. Go/no-go determination.

X. ORGANIZATIONAL COORDINATION CONSIDERATIONS (GENERIC / ILLUSTRATIVE)

Generic deconfliction logic for illustrative purposes only. This matrix does not reflect official agency authorities, which are governed by statute, regulation, and interagency agreements. Purpose: illustrate how lead agency ambiguity creates enforcement latency — the most common structural failure mode in multi-agency enforcement packages.

Enforcement Option	Generic Lead Lane	Generic Support Lane	Coordination Considerations	Approval Tier (Illustrative)
OSINT baseline + commercial AIS pull	Analytic coordination function	All consuming agencies	None — internal analytic process	Working level
§ 314(a) information sharing requests	Financial intelligence function	Financial enforcement coordination	None — standard process	Financial intelligence senior official
Internal preliminary review	Trade control + financial enforcement (parallel)	Intelligence coordination	None for internal action	Working level
Notice Letters to confirmed suppliers	Trade control function	Legal review; financial enforcement coordination	Diplomatic awareness as warranted	Trade control senior official
Registry de-flagging notifications	Diplomatic-maritime liaison function	Financial enforcement coordination	Interagency awareness	Senior diplomatic official (illustrative)
Entity List additions	Trade control (multilateral review process)	Financial enforcement; diplomatic; defense; energy reviewers	Diplomatic bilateral as warranted	Multilateral review committee consensus
Section 311 finding / special measures	Financial intelligence function (Treasury lead)	Legal review; diplomatic; policy coordination	Policy, legislative, and IC notification per statute	Senior Treasury official or designee
CISADA warning letters	Financial enforcement function	Diplomatic; financial intelligence coordination	Diplomatic bilateral engagement	Financial enforcement senior official
§ 981(k) asset substitution referral	Criminal enforcement function	Financial intelligence; financial enforcement coordination	Policy, diplomatic, and IC awareness	Senior criminal enforcement official; court order required
E.O. designation (Level 4 gate)	Financial enforcement function	Diplomatic; policy; IC; legal coordination	Full interagency; legislative awareness as warranted	Cabinet-level; full interagency coordination required
Maritime enforcement coordination (Level 4 gate)	Relevant maritime enforcement authorities	Financial enforcement; diplomatic; legal coordination	Full interagency; legislative awareness as warranted	Senior executive authority; applicable legal process required

Illustrative only. Actual authority, approval levels, and notification requirements governed by statute, regulation, and interagency agreements in effect at time of action.

XI. OBSERVABILITY METRICS AND ENFORCEMENT CADENCE

Three review tiers — weekly operational, monthly strategic, quarterly escalation gate. All data sources are commercial or open-source. No classified collection required for baseline monitoring. Metrics convert enforcement effectiveness from assertion to measurable, revisable output.

Weekly Operational Indicators

Indicator	Data Source	What Change Signals	Threshold for Escalation
AIS vessel churn at Node 7+9	Commercial AIS subscription	Network adaptation: new vessels replacing designated ones; dark period patterns	3+ new vessels/month OR >48hr dark period for known shadow fleet vessels → flag for Level 3 review
New flag registry filings for vessels previously calling Node 7	AIS + flag state public registries	Registry de-flagging pressure response; shadow fleet re-flagging	2+ re-flagging events in 30 days → escalate registry pressure
P&I club coverage changes for identified vessels	Lloyd's List Intelligence	P&I denial pressure taking effect; vessels losing coverage	Coverage loss for 2+ vessels → document as effectiveness indicator
Refinery flaring signature at Node 9	Sentinel-2 infrared (free, weekly pass)	Refinery operational tempo despite supply chain constraint	Sustained flaring despite Level 3 actions → satisfies predicate for CISADA escalation gate review

Monthly Strategic Indicators

Indicator	Data Source	What Change Signals	Threshold
Construction activity at Node 3	Sentinel-2 monthly; Google Earth; Planet commercial	Construction tempo maintaining vs. slowing; supply chain constraint taking effect	Continued construction tempo despite Level 2–3 actions → accelerate Entity List and FDPR
Supply chain substitution	Commercial trade databases (Panjiva, ImportGenius)	Network adaptation to constraint; new enforcement targets	Documented substitution via new country or entity → immediate § 314(a) and BIS review
Contractor presence at construction sites	Sentinel-2 change detection; road traffic analysis	Notice Letter constraint affecting contractor presence	Observable workforce reduction → document as effectiveness indicator
Financial SAR pattern changes	FinCEN SAR aggregation (internal access required)	Shadow financial flows adapting to Section 311 pressure; new intermediaries	New SAR clustering pattern → immediate § 314(a) queries for identified institutions

Quarterly Escalation Gate Review

Review Element	Assessment Question	Escalation Standard	Downshift Standard
Overall effectiveness	Are construction tempo, vessel operations, and financial flows de-risked relative to baseline?	Escalate if Level 2–3 actions produce no measurable change in SLRI indicators after 90 days	Downshift if indicators show >30% reduction in activity relative to baseline
Collateral risk review	Are enforcement actions producing unintended civilian impacts? GL scope adequate?	Escalation paused if collateral impact assessment identifies unacceptable civilian harm risk	Downshift from active enforcement to monitoring if collateral risk exceeds defined threshold

Node 3 window update	Is Node 3 still pre-operational? Has construction tempo changed?	Escalate if Node 3 within 6 months of estimated operational status	Downshift if construction stalled for 60+ days → maintain monitoring, reduce enforcement tempo
Coalition coordination	Are FATF and allied jurisdictions implementing parallel measures?	Escalate if key jurisdictions not implementing → escalate diplomatic engagement; consider FATF referral	Downshift if coalition implementing effectively → coordinate de-escalation timing

Escalation Review Checklist (Quarterly)

- ☐ Identity standards met per Annex C and Pull-Out 3 checklist for all active entity-linked actions
- ☐ Custody hooks documented in Annex B and logged in Annex D Exhibit Register for all evidentiary claims
- ☐ Administrative record complete and versioned per Annex D for any pending Section 311 or designation actions
- ☐ All active claims still aligned to their stated confidence labels (OBSERVED / ASSESSED / INFERRED)
- ☐ No INFERRED assessment has been elevated to ASSESSED without new corroborating evidence documented
- ☐ Collateral risk and GL scope reviewed before any new external filing

ANNEX B — DOCUMENTARY HOOKS FOR EVIDENTIARY RECORD

Use only for tracing and documentary proof. Do not infer illicit activity from hook presence alone. All actions gated per Section 4.5 decision tree and escalation gate standards.

Hook Type	What It Establishes	Minimum Use Standard	Log in Annex D
AIS vessel call record	Vessel presence at Node 7/9 general area; call frequency; flag registry; P&I club	Geographic bounding box match + vessel MMSI confirmed from AIS provider; date range and provider noted	Log with exhibit number, retrieval date, provider, query parameters
Export license application BIS SNAP-R	Supplier applied for controlled equipment export license to Iran or Iranian-linked entity	License number or application record confirmed from BIS public records; entity name match confirmed	Log with license number or application ID, date, BIS source
Shipping manifest	Physical shipment linkage between supplier and end-user	Bill of lading or manifest confirmed from commercial trade database or customs record; both party names confirmed	Log with manifest number, trade database source, date
SAR filing / § 314(a) response	Financial institution documented suspicious transaction activity linked to target entity	SAR or § 314(a) response received; entity linkage documented	Log with SAR reference or § 314(a) request ID, date, responding institution
Corporate registry extract	Legal entity name, jurisdiction, beneficial ownership, registered address	Registry extract dated within 12 months of action; source jurisdiction and registry noted	Log with registry name, jurisdiction, extract date, file reference
Payment documentation	Financial transaction linking two entities in the supply chain	Payment reference confirmed from bank records, trade finance, or financial disclosure; both parties named	Log with document type, date, parties, file reference

ANNEX C — IDENTITY STANDARDS FOR ENTITY-LINKED ACTIONS

All enforcement actions requiring entity identification must meet these standards before proceeding. Applies to Notice Letters, Entity List additions, Section 311 administrative records, and designation packages.

Element	Standard	Source Examples	Status Field
Legal entity name	Exact legal name as registered	Corporate registry extract; BIS SNAP-R application; shipping manifest	☐ Confirmed / ☐ Pending / ☐ Not found
Jurisdiction of incorporation	Country and registration authority confirmed	Corporate registry; commercial database (Dun & Bradstreet, ICIJ)	☐ Confirmed / ☐ Pending / ☐ Not found
Physical address	Street address confirmed — not PO box	Corporate registry; shipping manifest; export license application	☐ Confirmed / ☐ Pending / ☐ Not found
Beneficial ownership	Controlling interest ≥25% or equivalent threshold identified	Corporate registry; ICIJ Offshore Leaks; FinCEN BOI registry	☐ Confirmed / ☐ Pending / ☐ Not found
Transaction record	At least one independent transaction linking entity to relevant supply chain	Shipping manifest; export license; SAR; § 314(a) response	☐ Confirmed / ☐ Pending / ☐ Not found
SDN/Entity List cross-reference	Check at time of action	OFAC sanctionslist.ofac.treas.gov ; BIS bis.doc.gov	☐ Confirmed / ☐ Pending / ☐ Not found
End-user designation confirmation (Notice Letters only)	End-user designated on OFAC SDN List or BIS Entity List at time of notice delivery	OFAC SDN List; BIS Entity List	☐ Confirmed / ☐ Pending / ☐ Not found

ANNEX D — EXHIBIT REGISTER

Exhibit freeze/versioning rules: Each exhibit assigned unique number upon first entry (not reused). Exhibits frozen at retrieval — original file retained unmodified. If superseded, original retained and new exhibit number assigned for updated version.

Exhibit	Description	Source	Retrieval Date	Chain-of-Custody	Reproducibility Pointer
D-1	Late 2025 Sentinel-2 imagery — Node 3 construction progress	ESA Copernicus Sentinel-2	Oct–Nov 2025	Stored in secure OSINT repository; access log maintained	Tile query: Oct–Nov 2025, bands 4/8/11. Free at scihub.copernicus.eu
D-2	Sentinel-2 2022–2025 archive — construction sequence all nodes	ESA Copernicus Sentinel-2	2022–2025 archive	Full archive retained; query parameters documented	Time-series at node general coordinates, annual intervals, bands 4/8 + SWIR
D-3	Commercial AIS sample — Node 7/9 port call analysis	MarineTraffic or VesselFinder	To be compiled upon subscription	Raw extract retained; redacted copy for dissemination	AIS filter: bounding box Node 7+9 general areas, 24-month history
D-4	Google Earth historical imagery baseline all nodes	Google Earth Pro	Late 2025 most recent	Screenshot archive with date stamps	Historical imagery function at node general coordinates, earliest available through late 2025
D-5	BIS Entity List entries — identified supply chain entities	Commerce BIS public Entity List	Current at time of use	Public record	bis.doc.gov/index.php/policy-guidance/lists-of-parties-of-concern
D-6	OFAC SDN List entries — E.O. 13382 designated entities	OFAC SDN List	Current at time of use	Public record	sanctionslist.ofac.treas.gov filter E.O. 13382
D-7+	Add rows as compiled: shipping manifests, SAR patterns, § 314(a) responses, Lloyd's records, UN Panel reports	Per source	Per retrieval	Per source handling	Per source

APPENDIX E — WHAT WOULD CHANGE MY MIND

This appendix states explicitly what evidence would cause each key assessment in this paper to be revised — upward or downward. It is not a prediction. It is a falsifiability statement. A framework that cannot specify its own disconfirming conditions is advocacy, not analysis.

Assessment	Would Revise UPWARD if:	Would Revise DOWNWARD if:
H2: Resilience-oriented logistics enablement (ASSESSED; moderate confidence)	Documentary evidence of state or security-linked operators confirmed via corporate registry or shipping records Recurring vessel association with sanctioned maritime networks confirmed via AIS and designation cross-reference Financial intermediaries matching proliferation financing typologies identified via § 314(a) or SAR	Verified civilian permitting records from Iranian domestic authorities confirmed via open sources Transparent commercial operators with conventional market-facing counterparty records Absence of any designated-entity nexus after full identity resolution at Level 2–3
Node 3 function assessment (ASSESSED; moderate confidence)	Controlled goods procurement records linking Node 3 supply chain to export-controlled equipment categories Satellite imagery showing specialized installation consistent with a specific industrial function Financial flows consistent with a specific industrial production or assembly operation	Civilian industrial permitting records with identified operator and conventional supply chain Satellite imagery showing standard civilian manufacturing profile inconsistent with assessed function Identified operator with transparent, commercial, non-designated counterparties
Node 4 contents (ASSESSED; moderate confidence)	Delivery of controlled goods via adjacent port confirmed via cargo manifest or export license records Operator identity confirmed and linked to designated entity or controlled procurement network	Transparent civilian storage operator identified with conventional warehousing counterparties Contents confirmed as standard commercial inventory via open-source or diplomatic reporting
Node 3 timeline estimate (ASSESSED; moderate confidence, 12–18 months)	Accelerated construction tempo visible in monthly Sentinel-2 imagery Delivery of major equipment visible at site Workforce increase visible via site activity analysis	Construction activity ceases or slows significantly for 60+ consecutive days No major equipment deliveries visible over 90-day monitoring window Workforce reduction visible via site activity analysis
CPEC adjacency / secondary sanctions eligibility (INFERRED; lower confidence)	Financial intelligence confirming transactions between cluster-linked entities and CPEC-affiliated entities Shipping records linking cluster supply chain to Chinese state-affiliated vendors SAR filings or § 314(a) responses identifying Chinese financial intermediaries	Supply chain analysis confirms no Chinese state-affiliated entity involvement at any tier Financial intelligence confirms cluster financing exclusively from non-CPEC sources

The most important disconfirming indicator for the entire framework is one this product cannot supply: affirmative civilian end-use confirmation from a source with direct access. If that evidence exists, the framework's Level 2–3 enforcement eligibility assessments should be revisited before any action proceeds.